

WEEK 3 - ASSIGNMENT

Objective:

Use Subqueries, CTEs, and Window Functions to analyze sales data from the Superstore dataset.

Tasks

Step 1: Setup Data

1. Import the Superstore dataset into a table named **superstore_raw**.
2. Create these 3 tables from it:
 - a. customers
 - b. orders
 - c. products
3. Insert data into these tables using **SELECT DISTINCT**.

Step 2: Perform Required Queries

Write and execute SQL queries for each of the following:

1. Find all orders where sales are greater than the average sales. (*Subquery*)
2. Find the highest sales order for each customer. (*Subquery*)
3. Calculate total sales for each customer. (*CTE*)
4. Find customers whose total sales are above average. (*CTE + Subquery*)
5. Rank all customers based on total sales. (*Window Function*)
6. Assign row numbers to each order within a customer. (*Window Function + PARTITION BY*)
7. Display top 3 customers based on total sales. (*Window Function*)

Step 3: Final Combined Query

Write **one final query** that shows:

- Customer Name
- Total Sales
- Rank

(Use **JOIN + CTE + Window Function together**)

Mini Project: Customer Sales Insights

Answer the following using SQL:

1. Who are the top 5 customers?

```
9      SELECT
10      c.customer_name,
11      cs.total_sales,
12      RANK() OVER (ORDER BY cs.total_sales DESC) AS customer_rank
13  FROM customer_sales cs
14  JOIN customers c
15  ON cs.customer_id = c.customer_id
16  ORDER BY customer_rank;
```

Result Grid Filter Rows: Export: Wrap Cell Content:			
	customer_name	total_sales	customer_rank
▶	Sean Miller	50086.14	1
	Tamara Chand	38035.70	2
	Raymond Buch	30234.70	3
	Tom Ashbrook	29191.24	4
	Adrian Barton	28711.22	5
	Sanjit Chand	28284.68	6
	Ken Lonsdale	28143.84	7
	Hunter Lopez	25746.60	8
	Sanjit Engle	24418.88	9
	Christopher Conant	24258.16	10

2. Who are the bottom 5 customers?

Result Grid Filter Rows: Export: Wrap C		
	customer_id	total_sales
▶	TS-21085	9.68
	LD-16855	10.60
	MG-18205	24.64
	CJ-11875	33.04
	RS-19870	44.66

3. Which customers made only one order?

```

1 • SELECT
2     customer_id,
3     COUNT(order_id) AS total_orders
4 FROM orders
5 GROUP BY customer_id
6 HAVING COUNT(order_id) = 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
customer_id	total_orders			

4. Which customers have above-average sales?

```

1 • SELECT *
2 FROM orders
3 WHERE sales >
4 (
5     SELECT AVG(sales)
6     FROM orders
7 );

```

Result Grid

Filter Rows:

Export:

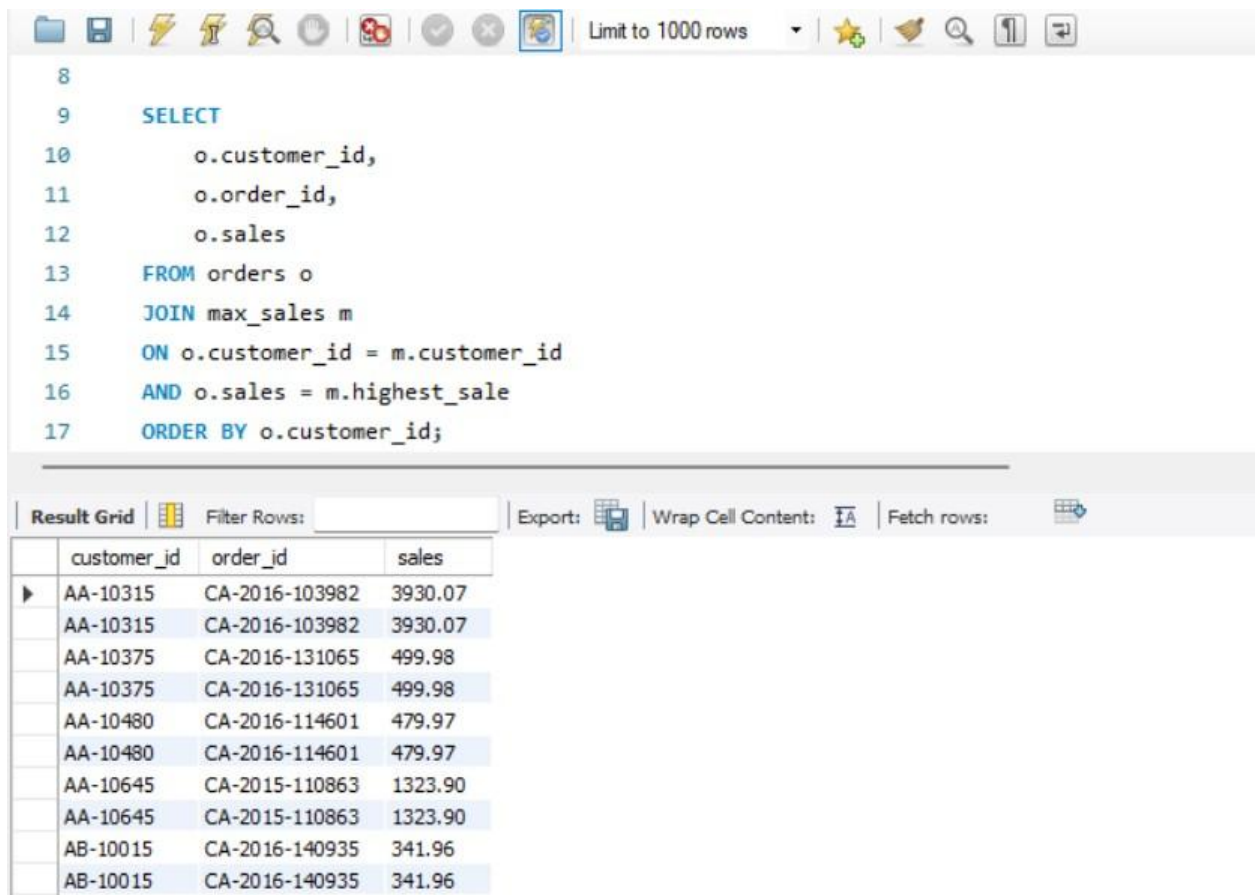
Wrap Cell Content:

Fetch rows:

	order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
▶	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0.00	41.91
	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0.00	219.58
	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	FUR-TA-10000577	957.58	5	0.45	-383.03
	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002275	907.15	6	0.20	90.72
	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	FUR-TA-10001539	1706.18	9	0.20	85.31
	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002033	911.42	4	0.20	68.36
	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	OFF-BI-10003656	407.98	3	0.20	132.59
	CA-2014-105893	2014-11-11	2014-11-18	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0.00	13.32
	CA-2015-106320	2015-09-25	2015-09-30	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0.00	240.26
	US-2015-150630	2015-09-17	2015-09-21	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.50	-1665.05
	CA-2016-117590	2016-12-08	2016-12-10	First Class	GH-14485	TEC-PH-10004977	1097.54	7	0.20	123.47
	CA-2015-117415	2015-12-27	2015-12-31	Standard Class	SN-20710	FUR-BO-10002545	532.40	3	0.32	-46.98

orders 21

5. What is the highest order value per customer?



```

8
9  SELECT
10      o.customer_id,
11      o.order_id,
12      o.sales
13  FROM orders o
14  JOIN max_sales m
15  ON o.customer_id = m.customer_id
16  AND o.sales = m.highest_sale
17  ORDER BY o.customer_id;

```

	customer_id	order_id	sales
▶	AA-10315	CA-2016-103982	3930.07
	AA-10315	CA-2016-103982	3930.07
	AA-10375	CA-2016-131065	499.98
	AA-10375	CA-2016-131065	499.98
	AA-10480	CA-2016-114601	479.97
	AA-10480	CA-2016-114601	479.97
	AA-10645	CA-2015-110863	1323.90
	AA-10645	CA-2015-110863	1323.90
	AB-10015	CA-2016-140935	341.96
	AB-10015	CA-2016-140935	341.96

WEEK 3 – ASSIGNMENT REPORT

Assignment Title

SQL Analysis using Subqueries, CTEs, and Window Functions on the Superstore Dataset

Student Details

Name: Palak Rathore

College: Dehradun Institute of Technology (DIT)

Program: B.Tech CSE (Full Stack & DevOps)

Internship: Celebal Technologies Internship Program 2026

Week: 3

Objective

The objective of this assignment was to analyze the Superstore sales dataset using advanced SQL concepts such as Subqueries, Common Table Expressions (CTEs), Window Functions, and JOIN operations. The assignment aimed to strengthen SQL skills by solving practical business-related problems and generating meaningful insights from sales data.

Step 1: Data Setup

Importing the Dataset

The Superstore dataset was imported into MySQL Workbench using the Table Data Import Wizard and stored in a table named **superstore_raw**. This table served as the source for all further analysis.

Creating Normalized Tables

To organize the data efficiently, three separate tables were created:

- **customers**
- **orders**
- **products**

This normalization reduced redundancy and improved the overall database structure.

Inserting Data

The data from the raw table was inserted into the newly created tables using the INSERT INTO ... SELECT DISTINCT statement. Using SELECT DISTINCT ensured that duplicate records were not inserted into the normalized tables.

Step 2: SQL Analysis

1. Orders with Sales Above Average (Subquery)

A subquery was used to calculate the average sales value of all orders. Orders with sales greater than this average were retrieved to identify high-value transactions.

Outcome

The query successfully identified all orders whose sales exceeded the overall average sales value.

2. Highest Sales Order for Each Customer (Subquery)

A subquery was used to determine the highest-value order placed by every customer.

Outcome

The result displayed the maximum sales transaction corresponding to each customer.

3. Total Sales per Customer (CTE)

A Common Table Expression (CTE) was created to calculate the total sales generated by each customer by grouping records based on Customer ID.

Outcome

The query produced a summarized view of customer-wise sales.

4. Customers with Above-Average Sales (CTE + Subquery)

The CTE containing customer sales totals was combined with a subquery to identify customers whose total sales were greater than the average customer sales.

Outcome

This query highlighted the customers contributing above-average revenue.

5. Ranking Customers (Window Function)

The RANK() window function was used to assign rankings to customers based on their total sales.

Outcome

Customers were ranked from highest to lowest according to their sales performance.

6. Row Number for Customer Orders

The ROW_NUMBER() function with the PARTITION BY clause was used to assign sequential numbers to each order placed by individual customers.

Outcome

Each customer's orders were numbered independently, making it easier to analyze their purchase history.

7. Top 3 Customers

Window Functions were used to identify the three customers with the highest total sales.

Outcome

The query returned the top-performing customers based on sales.

Step 3: Final Combined Query

A final SQL query was developed by combining **JOIN**, **CTE**, and **Window Functions**.

The query displayed:

- Customer Name
- Total Sales
- Customer Rank

Outcome

This provided a comprehensive summary of customer performance in a single result set.

Mini Project: Customer Sales Insights

The following business questions were answered using SQL:

- Identified the **Top 5 Customers** based on total sales.
 - Identified the **Bottom 5 Customers** based on total sales.
 - Found customers who placed **only one order**.
 - Identified customers whose **total sales were above average**.
 - Retrieved the **highest order value** for every customer.
-

Results

The assignment demonstrated the practical application of advanced SQL concepts to analyze sales data efficiently.

Some important observations include:

- High-value customers contributed a significant portion of the total sales.
- Customer rankings helped identify the most valuable customers.

- Several customers had only one recorded order, indicating potential opportunities for improving customer retention.
 - Window Functions simplified ranking and partitioning operations.
 - CTEs made complex queries more readable and easier to manage.
-

Challenges Faced

While completing the assignment, I encountered a few practical challenges. Initially, I faced duplicate key issues while creating the normalized tables because some Product IDs appeared multiple times in the dataset. I resolved this by modifying the table structure appropriately. I also spent time understanding how CTEs and Window Functions work together, especially when combining them with JOIN operations to generate the final report.

Key Learnings

Through this assignment, I gained practical experience in:

- Importing and organizing datasets in MySQL.
 - Designing normalized database tables.
 - Using SELECT DISTINCT to populate tables.
 - Writing Subqueries for comparative analysis.
 - Using CTEs to simplify complex SQL logic.
 - Applying Window Functions like RANK() and ROW_NUMBER().
 - Combining multiple SQL concepts to solve business problems.
 - Organizing SQL projects professionally using GitHub.
-

Conclusion

This assignment provided valuable hands-on experience in advanced SQL and data analysis. It helped me understand how different SQL techniques can be combined to answer real-world business questions efficiently. In addition to improving my SQL skills, I also gained experience in organizing project files, documenting my work, and maintaining a structured GitHub repository, which are important practices in real-world data engineering projects.